



WHAT MAKES AN APP SUCCESSFUL?

A MACHINE LEARNING
ANALYSIS OF THE GOOGLE
PLAY STORE

presented by

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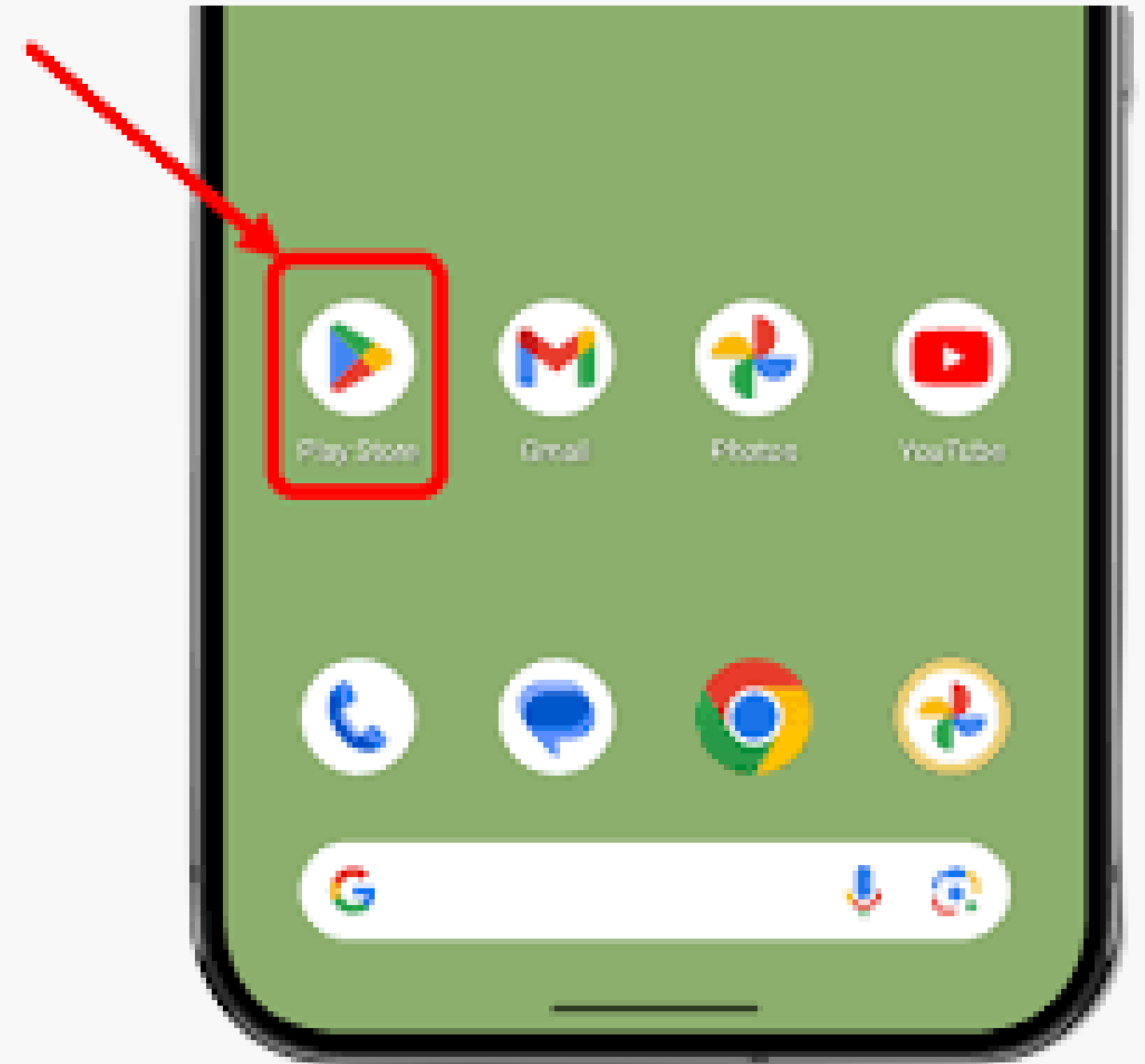
INTRODUCTION

Project Motivation

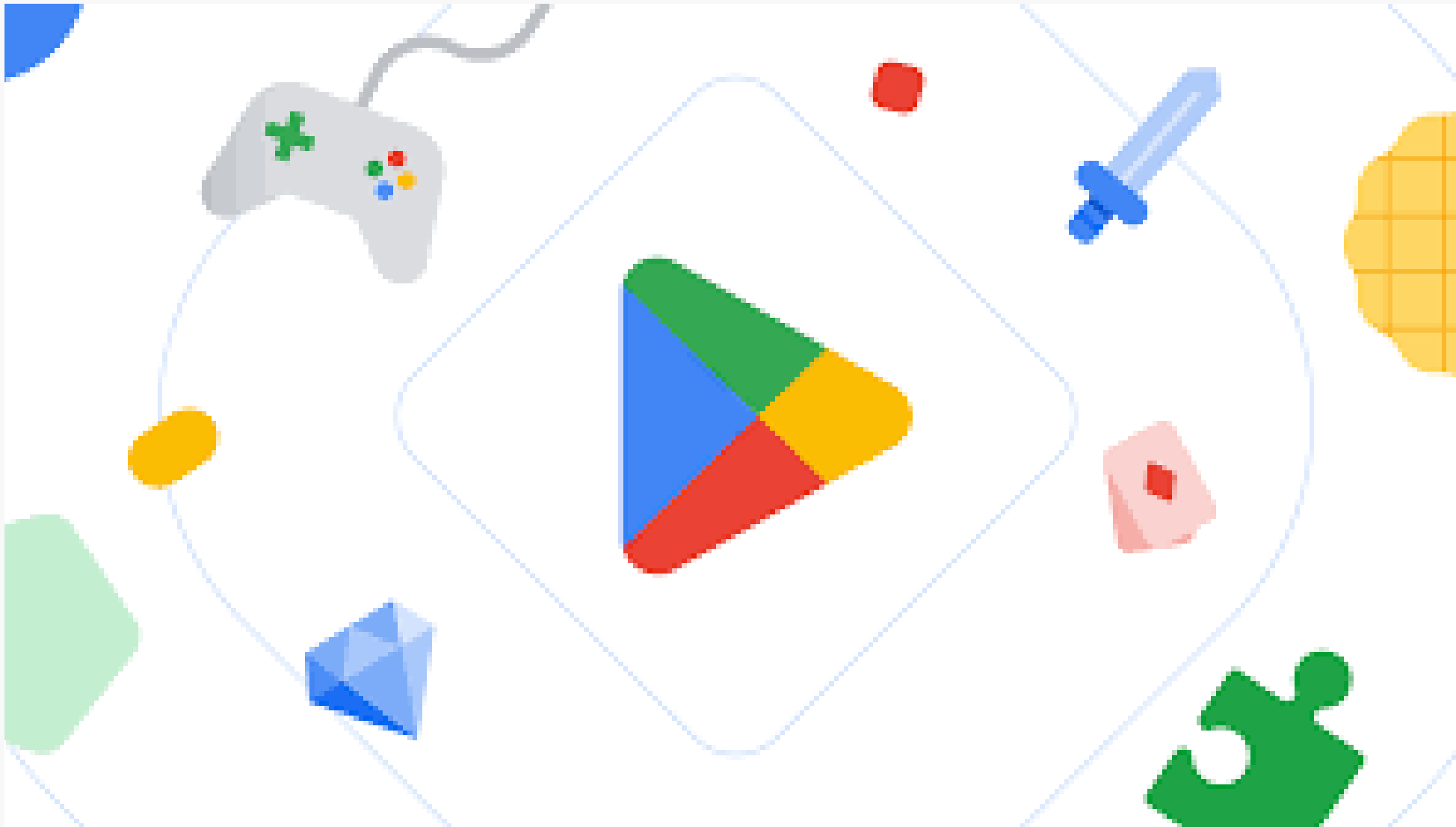
- The Google Play Store contains millions of applications across many categories.
- User engagement, ratings, and customer sentiment may influence app success.
- Businesses and developers can use analytics to better understand what drives application performance.

Project Goal

The goal of this project was to analyze factors associated with successful applications and build machine learning models capable of predicting app success.



RESEARCH QUESTIONS



1. Do free and paid apps differ in ratings?
2. Are reviews related to app ratings?
3. Do app ratings differ across categories?
4. Which variables best predict app success?
5. Can machine learning models accurately classify high-performing applications?

DATASET OVERVIEW



Dataset used

Google Play Store Dataset

Included:

- Ratings
- Reviews
- Installs
- Price
- App Type
- Categories

User Reviews Dataset

Included:

- Sentiment polarity
- Sentiment subjectivity
- Positive reviews
- Negative reviews
- Neutral reviews

Final Dataset

- 8,196 applications
- 19 variables

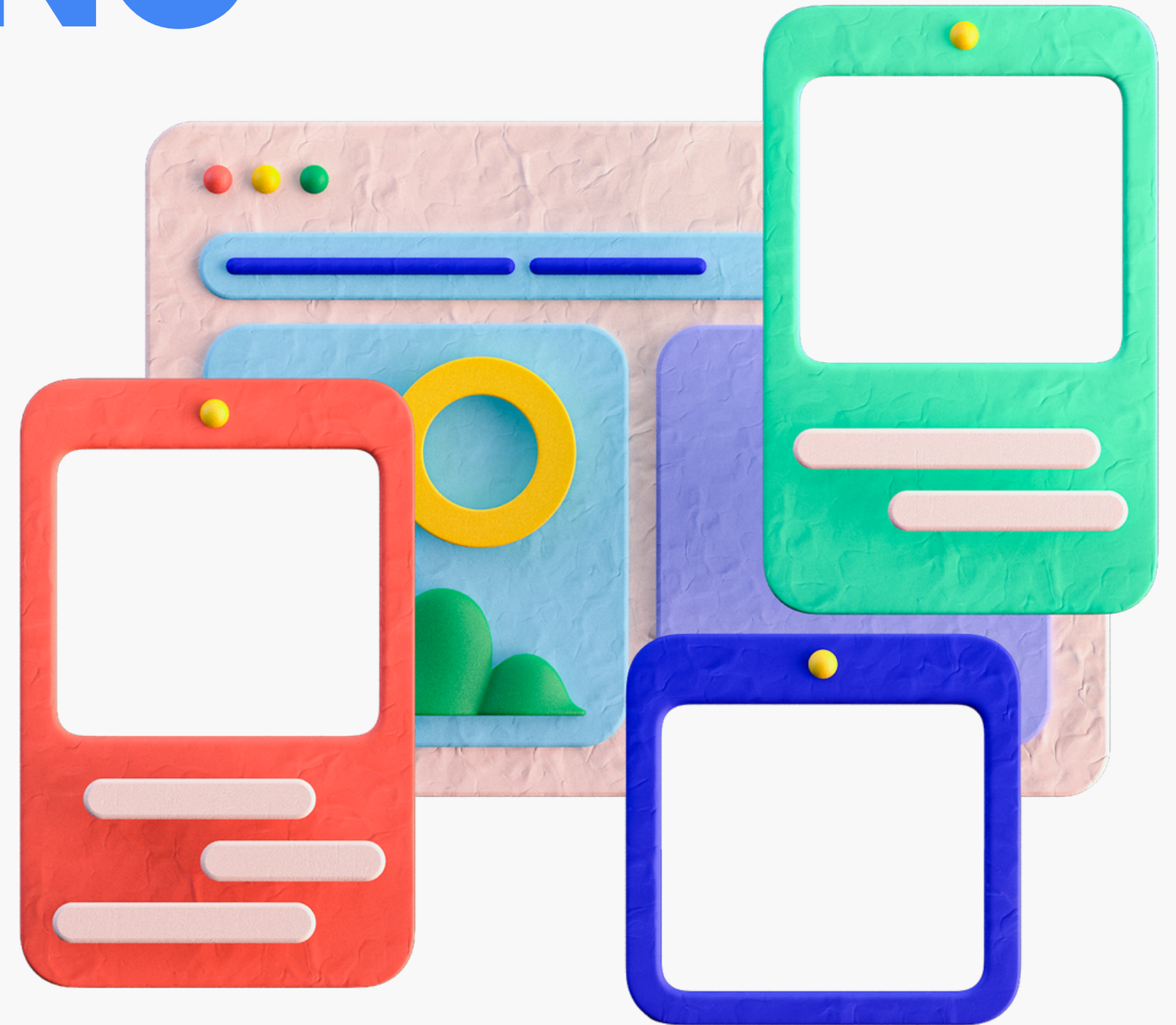
DATA CLEANING & PREPROCESSING

Data Cleaning Steps

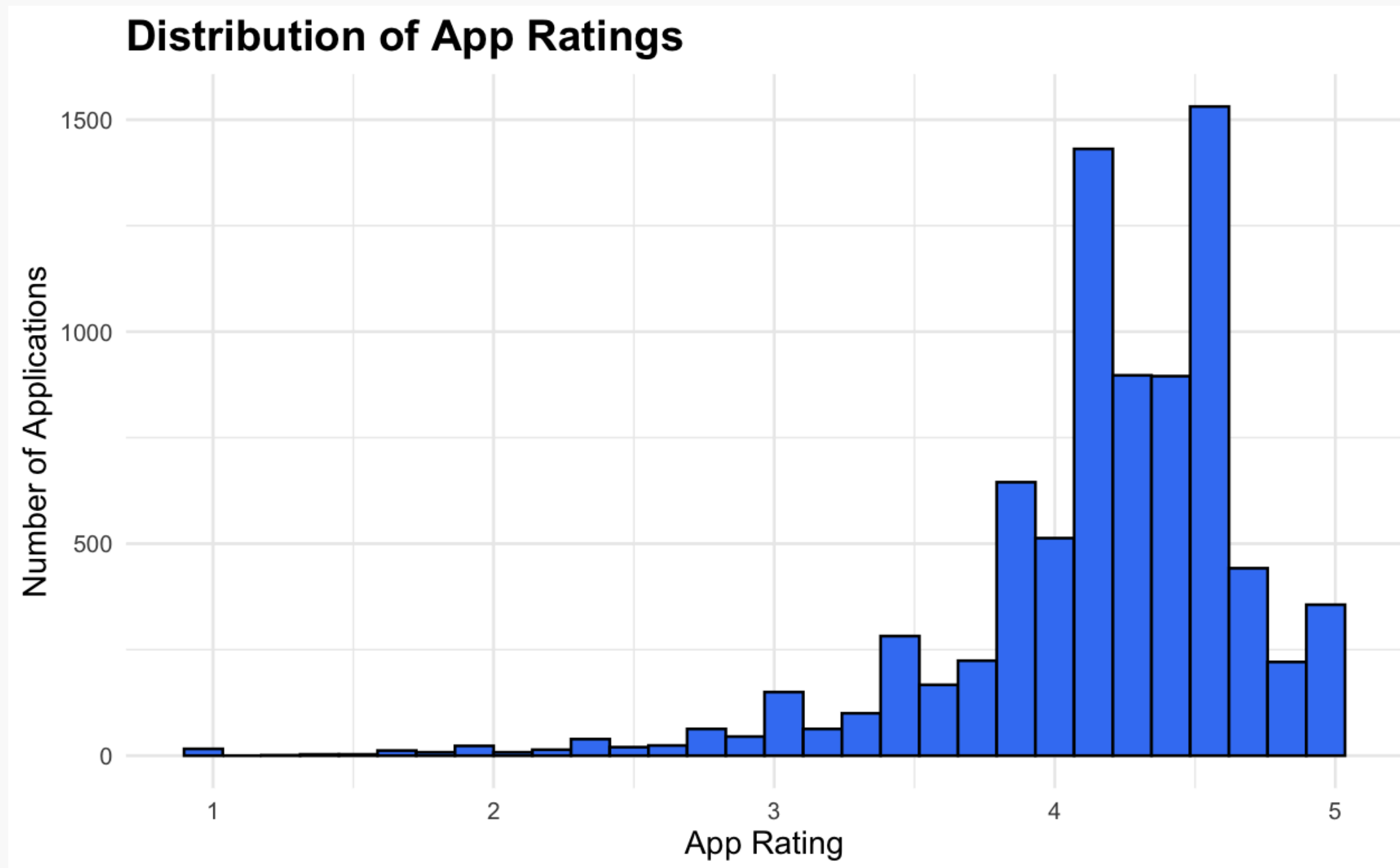
- Removed missing ratings
- Removed duplicate applications
- Converted variables to numeric
- Converted categorical variables into factors
- Created High_Performance classification variable
- Merged sentiment data with app data
- Removed corrupted category values

High Performance Definition:

1M+ installs



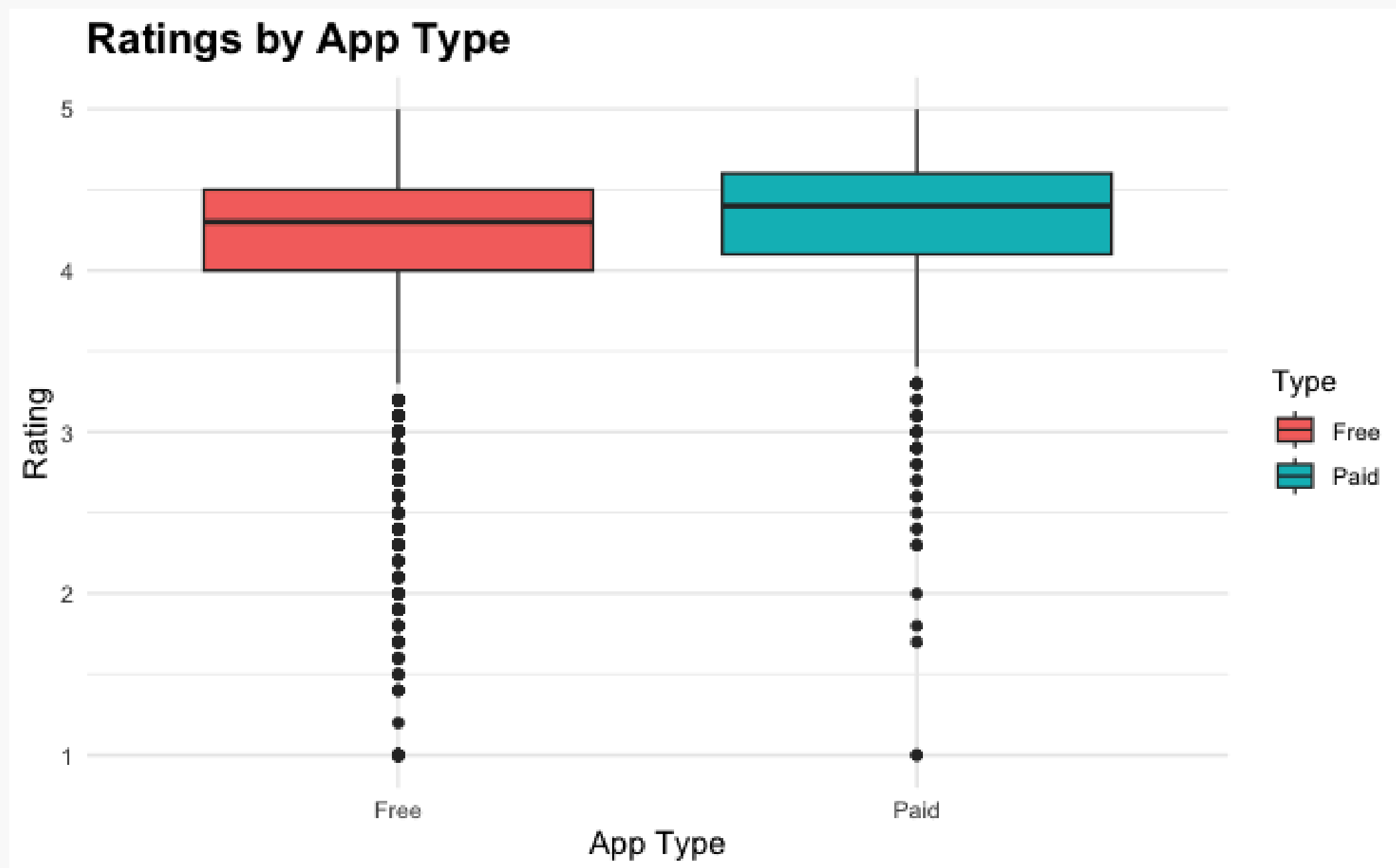
DISTRIBUTION OF APP RATINGS



Key Insights

- Most apps were rated between 4.0 and 4.7
- Low-rated apps were uncommon
- User satisfaction trends were generally positive

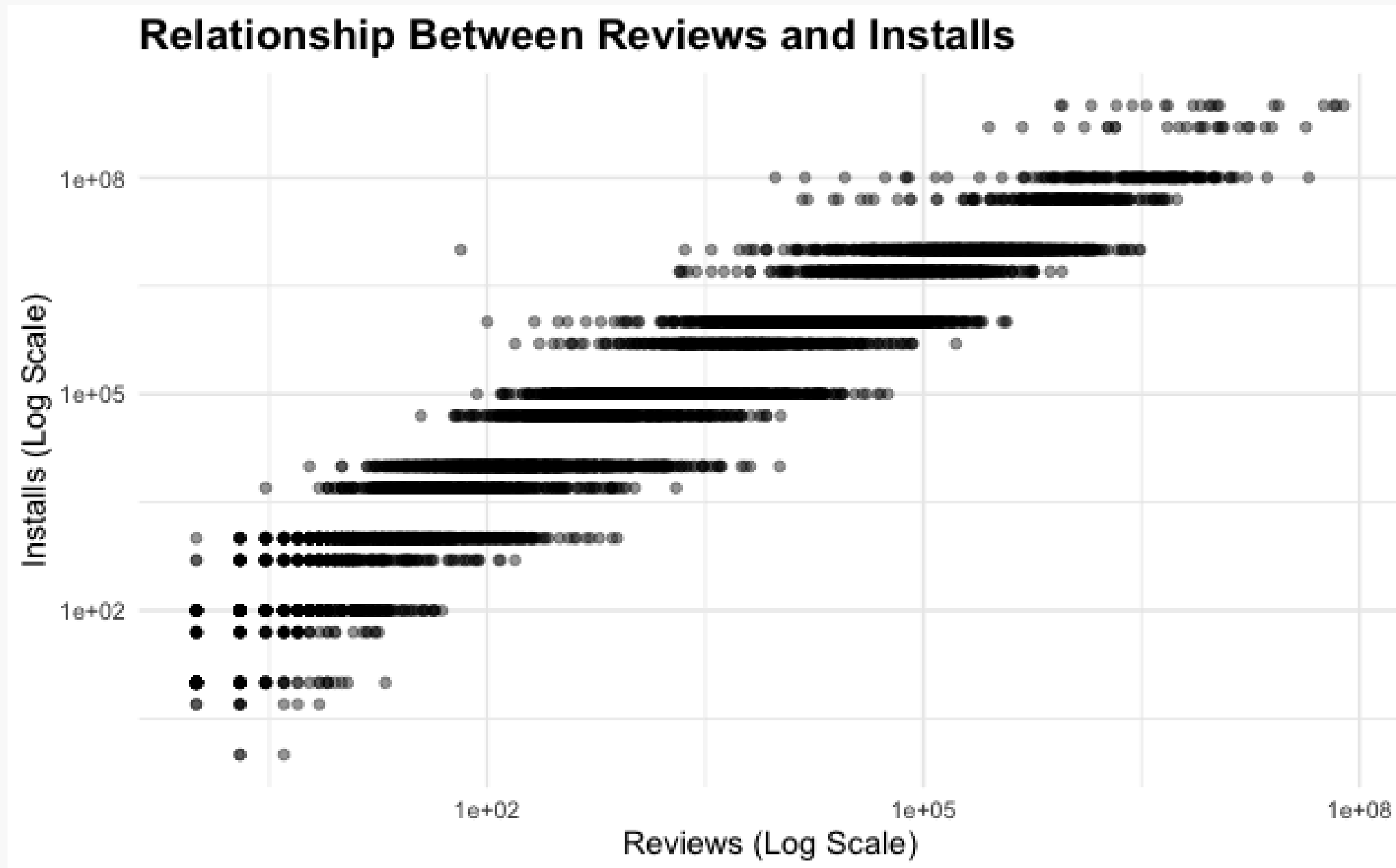
RATINGS BY APP TYPE



Key Insights

- Paid apps had slightly higher median ratings
- Free apps showed greater variability
- Both groups generally had ratings above 4.0

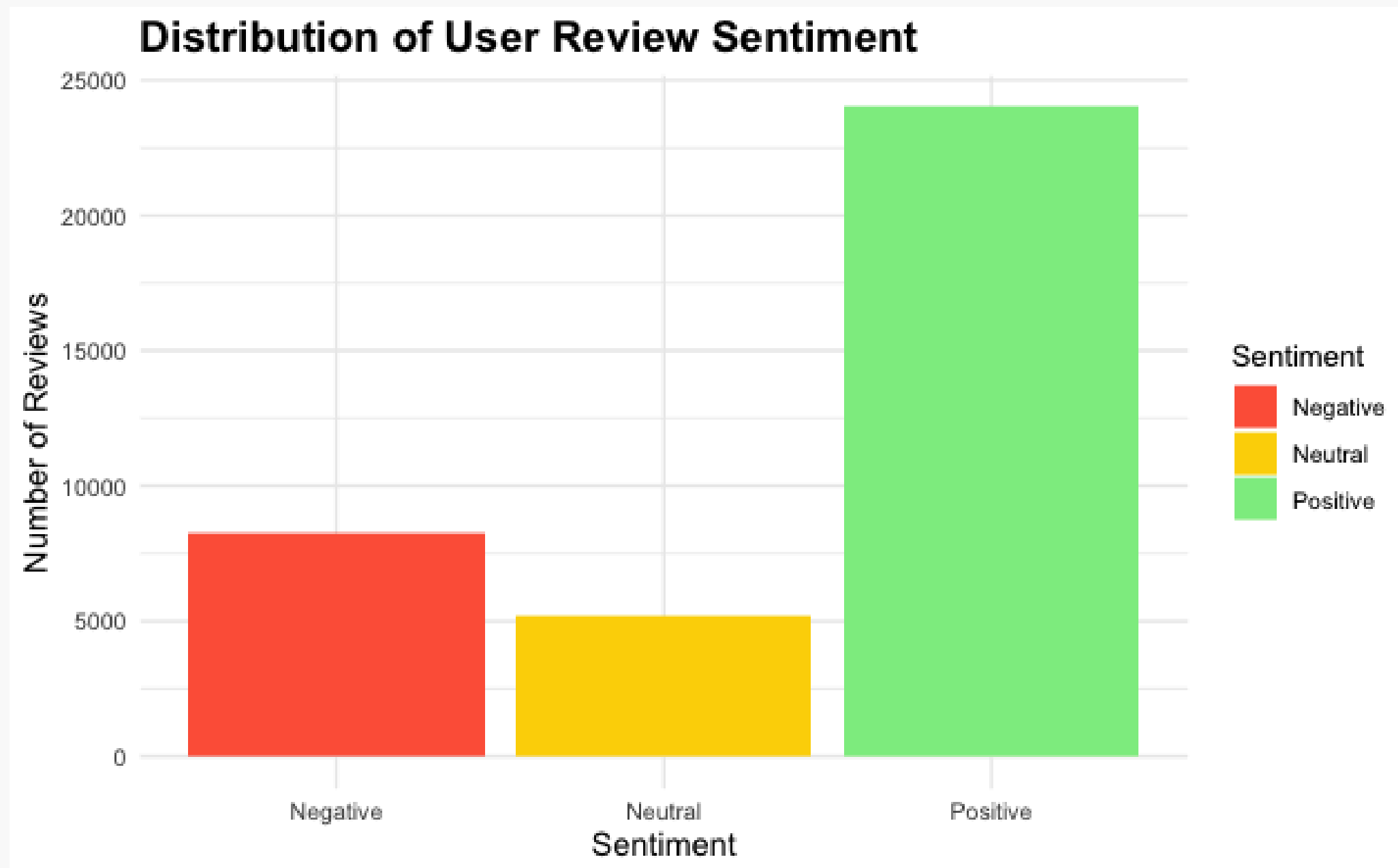
REVIEWS VS INSTALLS



Key Insights

- Strong positive relationship
- More reviews → more installs
- Engagement strongly related to app popularity

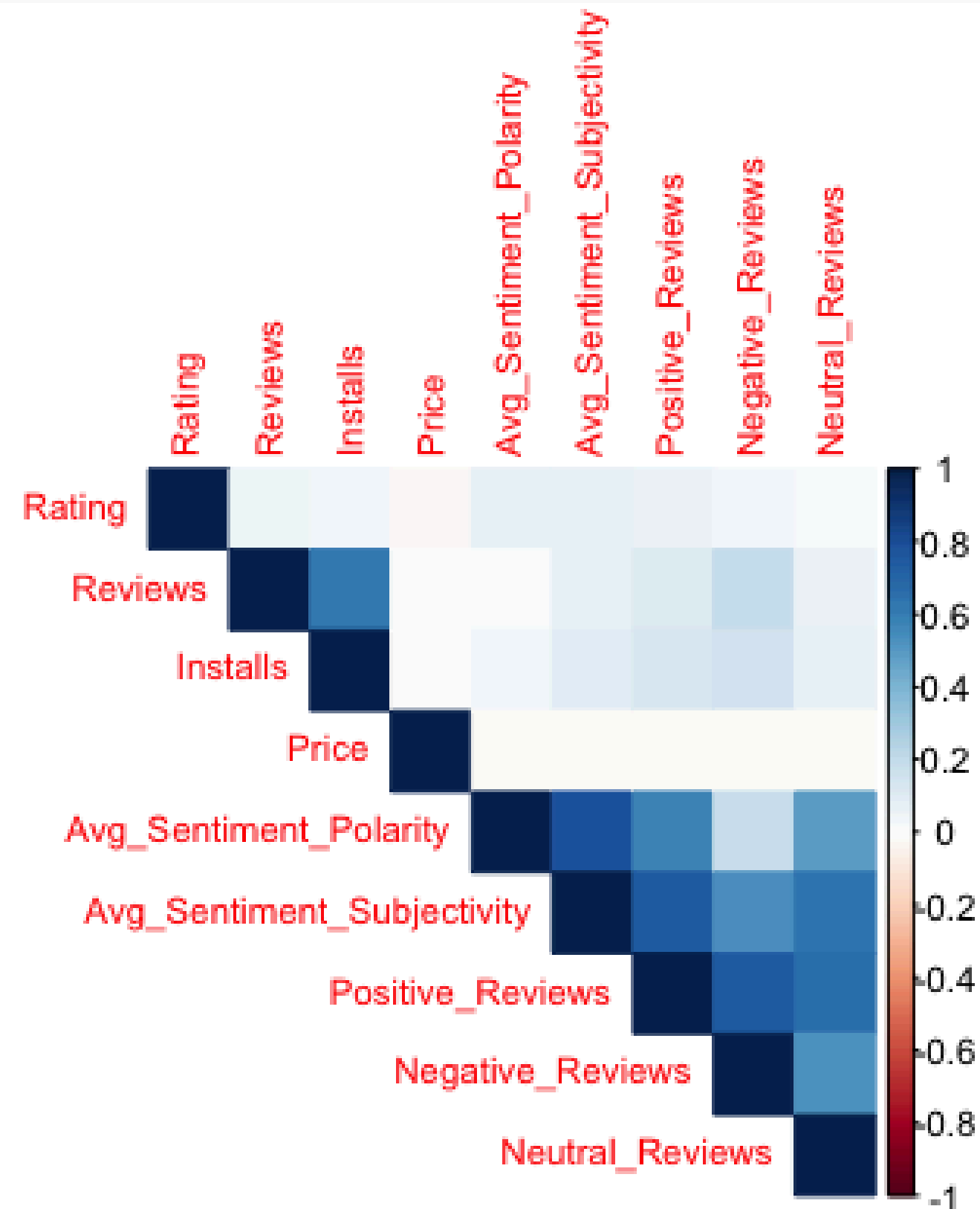
DISTRIBUTION OF USER REVIEW SENTIMENT



Key Insights

- Positive reviews dominated
- Negative reviews were much lower
- Overall customer sentiment was favorable

CORRELATION HEATMAP



Key Insights

- Reviews and installs had a strong positive relationship
- Sentiment variables were strongly related
- Price had weak relationships with most variables

HYPOTHESIS TESTING

Test	Result
T-Test	Paid apps had significantly higher ratings than free apps
Correlation Test	Reviews and ratings had a weak positive relationship
ANOVA	Ratings differed by category

**All tests were statistically significant
($p < 0.05$)**

LOGISTIC REGRESSION

Predictors Used

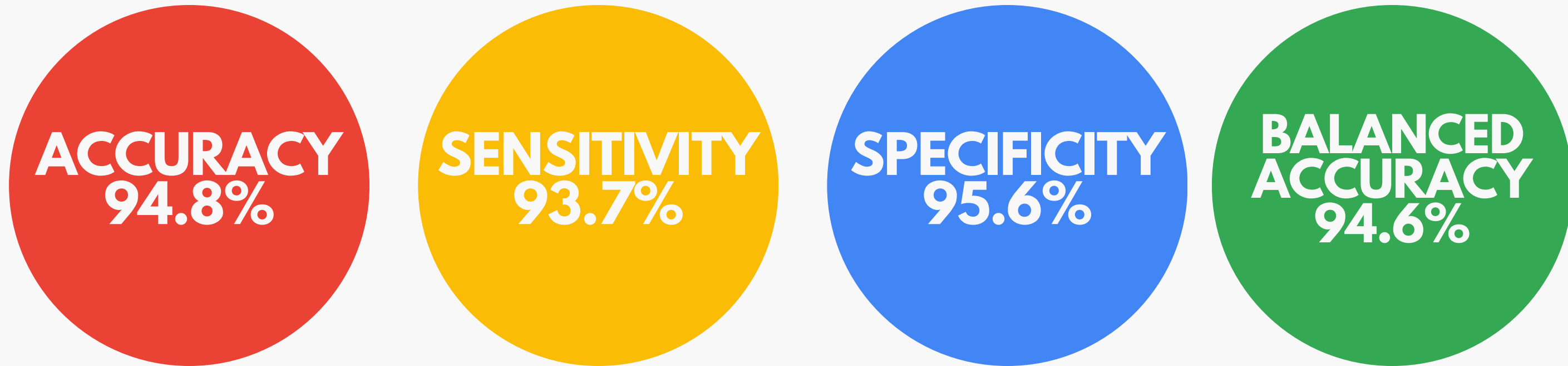
- Rating
- Reviews
- Price
- Type
- Sentiment variables

ACCURACY
93.5%

SENSITIVITY
87.9%

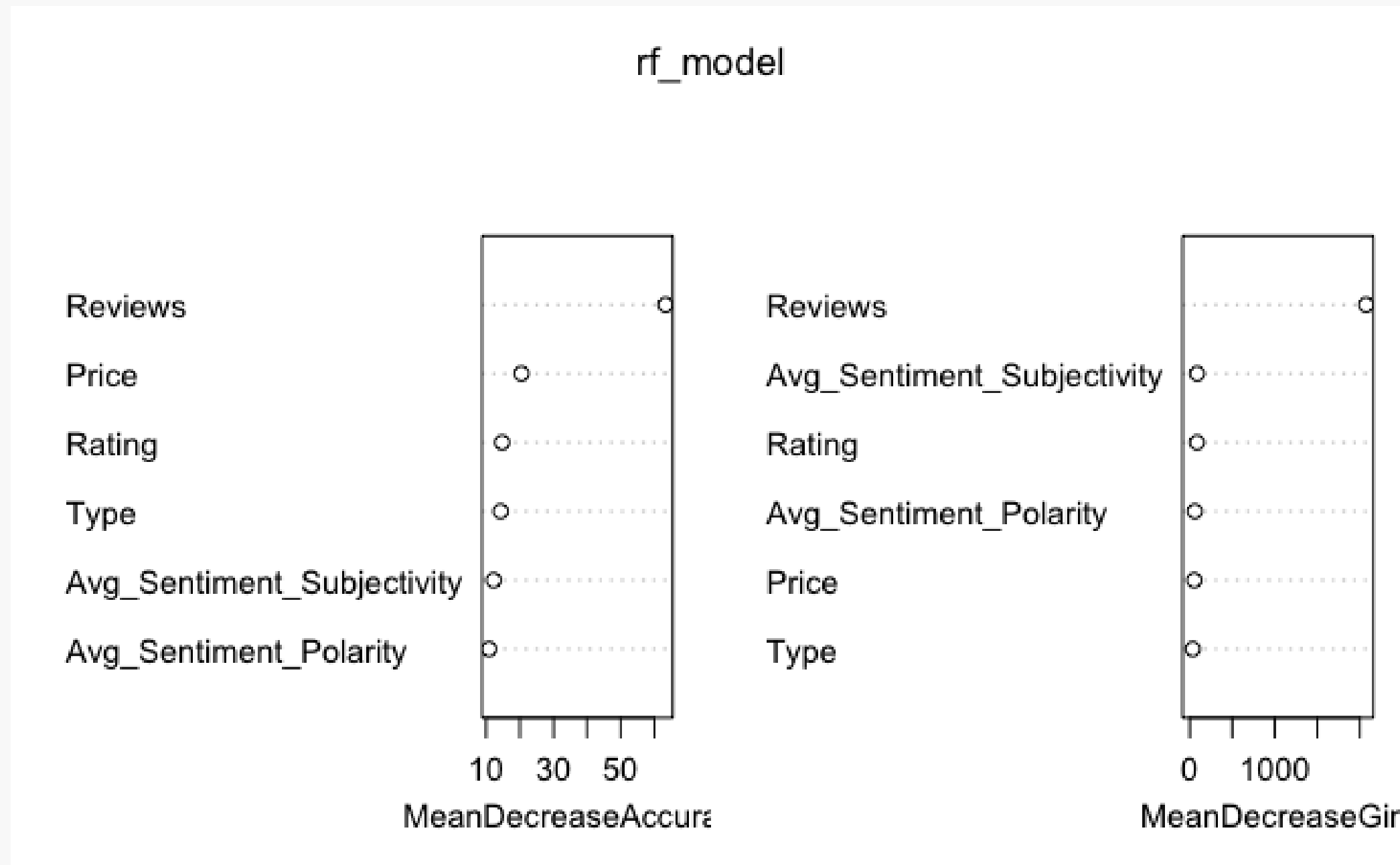
SPECIFICITY
97.4%

RANDOM FOREST



RANDOM FOREST OUTPERFORMED LOGISTIC REGRESSION

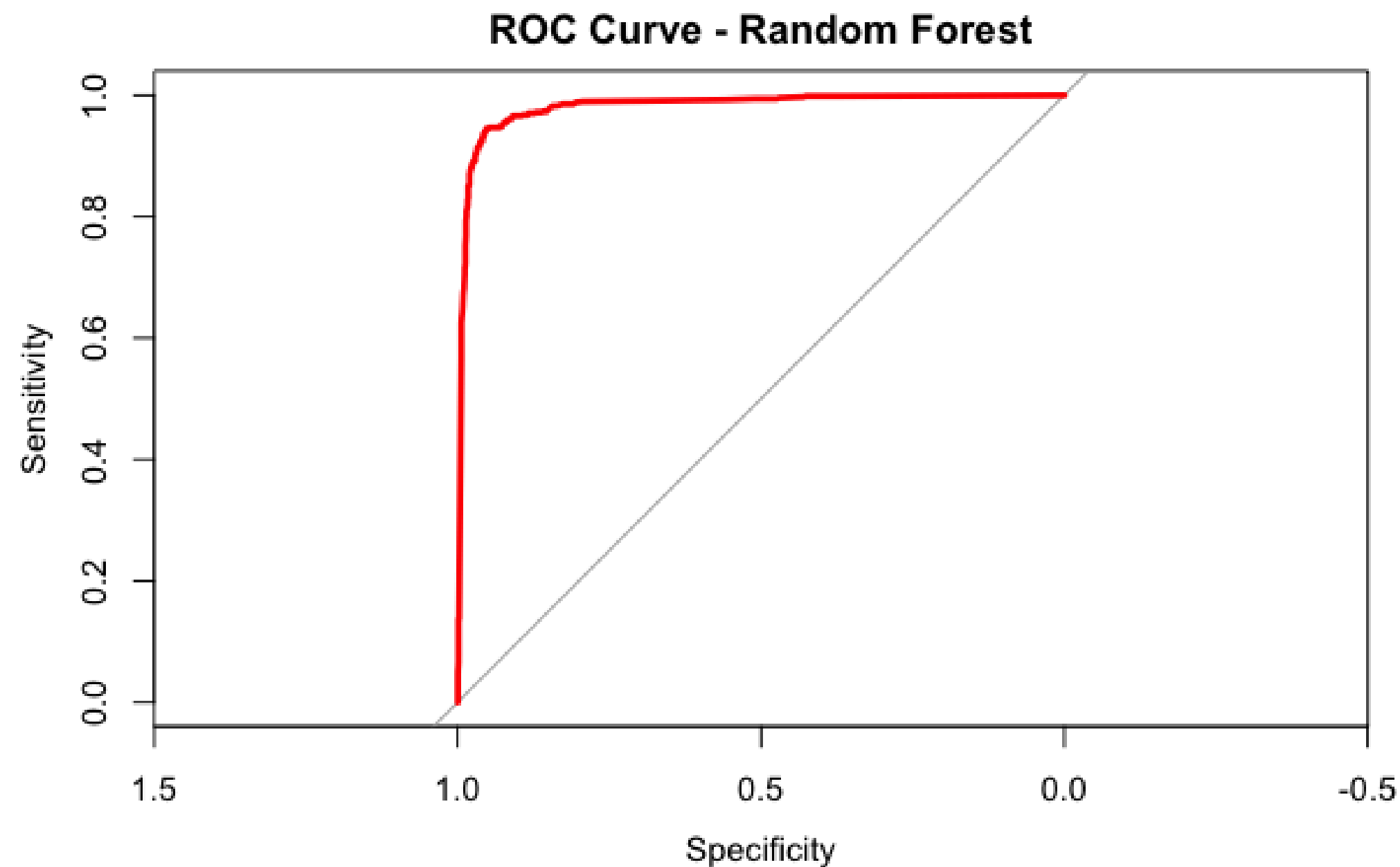
VARIABLE IMPORTANCE



Top Predictors

1. Reviews
2. Sentiment Subjectivity
3. Ratings

ROC CURVE & AUC



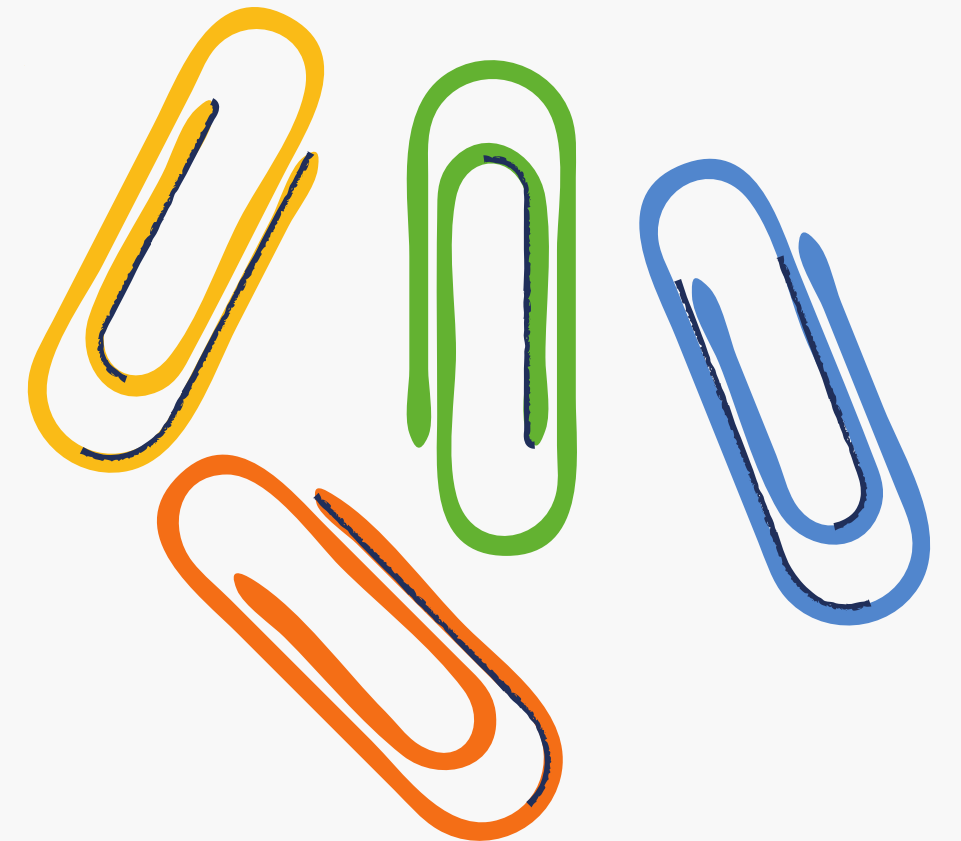
AUC Score

0.982

The model demonstrated
excellent classification
ability

KEY BUSINESS INSIGHTS

- Customer engagement strongly predicts app success
- Positive customer sentiment is associated with high-performing apps
- Paid apps received slightly higher ratings
- App category influences customer satisfaction
- Random Forest provided the strongest predictive performance



CONCLUSION

- Successfully analyzed Google Play Store application data
- Combined EDA, sentiment analysis, statistical testing, and machine learning
- Random Forest achieved the strongest predictive performance
- Reviews and sentiment variables were among the most important predictors

APPLICATION SUCCESS CAN BE EFFECTIVELY PREDICTED USING CUSTOMER ENGAGEMENT, SENTIMENT ANALYSIS, AND APP CHARACTERISTICS.



<https://www.kaggle.com/datasets/lava18/google-play-store-apps>